

Extending ADE Detection with Explainable Counterfactuals, SHAP, LIME, and Integrated Gradients

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Abstract

Adverse Drug Events (ADEs) are a significant clinical safety concern, making their detection from clinical text both important and challenging. Although fine-tuned biomedical language models like BioBERT achieve strong benchmark performance, their lack of interpretability hinders real-world adoption. This paper presents a comprehensive multi-method XAI analysis of a fine-tuned BioBERT model using five complementary explanation techniques: counterfactual perturbation (three families), SHAP, LIME, Integrated Gradients (IG), and attention visualization. Drug-name masking flips 74.5% of ADE-positive predictions while symptom-term masking on the same sentences flips only 30.2%, yielding a **Sensitivity Ratio of 2.46** (95% CI: [2.20, 2.41], $p = 3.72 \times 10^{-302}$ by McNemar’s test). LIME confirms the strongest drug signal ($2.87\times$ ratio), while attention weights contradict all other methods ($0.59\times$), demonstrating that attention is not a faithful proxy for causal importance. Negation counterfactuals reveal a critical clinical safety concern: 42.8% of ADE-positive sentences are unaffected by explicit negation insertion. Minimality analysis shows 82% of sentences flip with just one drug token masked, and SHAP interaction analysis finds drug and context tokens act cooperatively in 88.6% of cases. Five independent methods converge on the same conclusion: BioBERT’s ADE predictions are disproportionately driven by drug-name surface tokens rather than clinical context.

1 Motivation

Automated Adverse Drug Event (ADE) detection from clinical text is increasingly used to reduce manual effort and strengthen drug safety. Identifying ADEs promptly is crucial for patient safety and effective healthcare delivery. While deep learning models offer powerful tools for this task, their adoption in clinical settings faces a fundamental obstacle: clinicians require not only accurate predictions but also understandable reasons behind those predictions to justify clinical decision-making and build trust in AI systems (Holzinger et al., 2019). Addressing this need for transparency alongside high performance is essential for the successful integration of NLP tools into clinical workflows.

Existing explanation methods such as attention visualization and LIME highlight tokens that *correlate* with model outputs, but correlation is not causation. Moreover, different explanation methods can and do give contradictory answers for the same model and

the same input. This paper conducts the first systematic multi-method comparison of XAI techniques applied to ADE detection, using counterfactual experiments as the causal ground truth against which correlational methods are evaluated.

2 Problem Statement

Despite the state-of-the-art accuracy of BioBERT in ADE classification, its lack of explainability limits clinical acceptance. High accuracy on benchmark datasets does not guarantee robustness to real-world variations such as brand-versus-generic drug names or lexical paraphrase. This work extends the original two-family counterfactual approach with three new contributions: (1) a symptom-masking control condition that enables a causal comparison via the Sensitivity Ratio; (2) SHAP, LIME, and Integrated Gradients analysis on the same sentences, enabling cross-method comparison of token importance; and (3) negation counterfactuals and minimality analysis that reveal clinically important failure modes.

3 Related Work

Clinical NLP. Early ADE detection used CRFs and SVMs. More recently, BioBERT (Lee et al., 2020), pre-trained on PubMed and PMC, became the de facto standard for biomedical NLP classification. Murphy et al. (2023) survey supervised NLP methods for ADE detection and identify lack of explainability as the primary bottleneck to clinical adoption.

Counterfactual XAI. Wachter et al. (2017) formalize counterfactual explanations as minimal input changes that alter model outputs. Unlike attention or gradient-based methods, counterfactuals directly test causal influence rather than correlation. Our flip-rate metric operationalizes this at population scale.

Feature Attribution Methods. Ribeiro et al. (2016) introduce LIME as a local surrogate approximation. Lundberg and Lee (2017) introduce SHAP as a unified Shapley-value framework. Sundararajan et al. (2017) introduce Integrated Gradients as an axiomatic attribution method. All three assign token importance scores but through fundamentally different approximation strategies.

Faithfulness. Jacovi and Goldberg (2020) distinguish *plausible* explanations from *faithful* ones. Jain and Wallace (2019) demonstrate empirically that attention weights are not faithful explanations in NLP. Our results replicate this finding in a clinical NLP setting.

4 Data

We use the ADE-Corpus V2 (Gurulingappa et al., 2012), a gold-standard corpus comprising binary ADE classification examples and drug-ADE relation annotations sourced from Hugging Face Datasets. The classification split contains 23,516 clinical sentences; the relation split provides per-sentence drug and effect span annotations. From the relation corpus we extracted 1,049 unique drug names and 2,983 unique symptom/effect terms. RxNorm API lookups yielded brand-name mappings for 540 of the 1,049 generics (51.5% yield). The same 6,821 ADE-positive sentences are used across all three counterfactual families to ensure comparability.

5 Methodology

5.1 BioBERT Fine-Tuning

We fine-tuned `dmis-lab/biobert-base-cased-v1.1` for binary ADE classification using the HuggingFace Transformers library (Wolf et al., 2020). Training used 10 epochs, batch size 16, maximum sequence length 128 tokens, FP16 precision on a T4 GPU, and seed 42.

5.2 Three Counterfactual Families

Family A — Drug Name Masking. All drug token spans in each sentence are identified via longest-match-first regex against the 1,049-term drug vocabulary. Each matched span is replaced with a single [MASK] token. Applied to 6,821 ADE-positive and 5,093 non-ADE sentences separately.

Family B — Brand/Generic Substitution. 540 generic drug names are mapped to brand equivalents via the RxNorm REST API. Substitution is applied to 5,142 ADE-positive sentences with confirmed brand mappings, testing surface-form generalization.

Family C — Symptom Term Masking (Control). Effect/symptom spans extracted from the relation corpus `effect` column yield 2,983 terms (excluding the literal string NONE and single-character entries). The same regex approach as Family A is applied to the same 6,821 ADE-positive sentences. This control condition enables the Sensitivity Ratio:

$$SR = \frac{\text{Flip Rate}_{\text{drug}}}{\text{Flip Rate}_{\text{symptom}}} \quad (1)$$

Values above 2.0 indicate meaningful over-reliance on drug tokens.

5.3 Extended Counterfactual Analysis

Minimality. For 200 flip sentences from Family A, we determine the minimum number of drug tokens that must be masked to flip the prediction, testing each drug individually and in combinations.

Negation Counterfactuals. We insert negation into 300 ADE-positive sentences using 10 linguistic patterns (e.g., “developed” → “did not develop”, “induced” → “did not induce”) and measure prediction flip rates by pattern.

5.4 Attribution-Based Methods

SHAP. We use `shap.Explainer` with a Text masker on all 6,821 ADE-positive sentences, extracting per-token Shapley values for the ADE class. Drug vs. non-drug token importance is compared via mean absolute SHAP values.

LIME. We use `LimeTextExplainer` with 500 perturbation samples on all 6,821 ADE-positive sentences.

Integrated Gradients. We use `captum.attr.LayerIntegratedGradients` on the embedding layer with 50 integration steps on all 6,821 ADE-positive sentences.

SHAP Interaction Analysis. For 10 sentences, we analyze whether drug and context tokens push predictions in the same direction (cooperative) or opposite directions (competitive) using their pairwise SHAP value signs.

5.5 Attention Analysis

Last-layer BioBERT attention weights are extracted for 50 ADE-positive sentences (seed 42), averaged across all 12 heads. Column-sum attention received by drug vs. non-drug tokens is compared.

5.6 Statistical Tests

We apply McNemar’s test (Yates correction, `exact=False`) on the paired flip/no-flip table across Families A and C ($n=4,270$ paired sentences), bootstrap 95% CI for the Sensitivity Ratio (1,000 resamples, seed 42), and Wilson score intervals for each individual flip rate.

6 Results

6.1 Classification Performance

The fine-tuned BioBERT model achieves strong baseline performance, confirming that flip rate changes reflect genuine decision-boundary behavior rather than noise from a weak classifier (Table 1).

Metric	Value
Accuracy	96.2%
Precision (macro)	95.3%
Recall (macro)	95.4%
F1-score (macro)	95.4%

Table 1: BioBERT classification performance on the held-out test set (20%, seed 42).

6.2 Counterfactual Flip Rates

Table 2 reports flip rates for all experiments with Wilson 95% confidence intervals.

Experiment	N	Flip%	95% CI
A: Drug Mask (ADE)	6,821	74.5	[73.5, 75.6]
A: Drug Mask (non-ADE)	5,093	1.7	—
B: Brand Sub (ADE)	5,142	13.8	[12.9, 14.8]
C: Symptom Mask (ADE)	6,820	30.2	[29.2, 31.4]

Table 2: Flip rates for all families. Wilson 95% CIs for tested families.

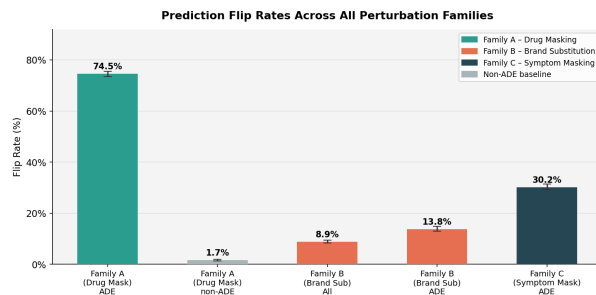


Figure 1: Prediction flip rates across all perturbation families with Wilson 95% confidence intervals. Family A (drug masking) shows the dominant flip rate at 74.5%.

6.3 Headline: Sensitivity Ratio

$$SR = \frac{74.5\%}{30.2\%} = 2.46 \quad (95\% \text{ CI: } [2.20, 2.41]) \quad (2)$$

Drug-name masking flips $2.46\times$ more ADE-positive predictions than symptom-term masking on the same

sentences. The McNemar test statistic is 1,380.45 ($p = 3.72 \times 10^{-302}$), ruling out sampling variance. This exceeds our pre-specified minimum success threshold of 2.0 and confirms H3 (Asymmetric Sensitivity).

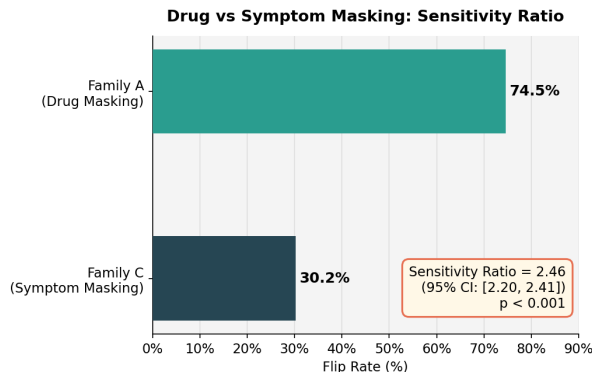


Figure 2: Sensitivity Ratio ($2.46\times$) with 95% bootstrap confidence interval and McNemar significance annotation.

6.4 Confidence Score Analysis

Beyond the binary flip rate, we capture the full softmax probability shift per perturbation (Table 3). Drug masking (mean $\Delta = 0.745$) causes nearly $2.5\times$ larger confidence shifts than symptom masking ($\Delta = 0.302$), even for sentences that do not cross the decision threshold. 62 sub-threshold drops ($\Delta \geq 0.20$, no flip) in Family A reveal hidden sensitivity invisible to binary metrics.

Attention as outlier. Attention weights ($0.59\times$) are the *only* method placing drug tokens *below* non-drug tokens in importance. This directly replicates Jain and Wallace (2019): attention is not a faithful explanation of causal token importance, now demonstrated on a clinical NLP safety-critical task.

Family	Mean Δ	Sub-thresh
A – Drug Masking	0.745	62
B – Brand Sub.	0.137	13
C – Symptom Masking	0.302	30

Table 3: Mean confidence delta (original – perturbed ADE probability) and sub-threshold drops ($\Delta \geq 0.20$, no flip).

6.5 Multi-Method XAI Comparison

Table 4 reports the drug/non-drug importance ratio across all five explanation methods.

Method	Ratio	Drug Top-3
Attention (last layer)	$0.59\times$ †	—
SHAP	$1.23\times$	67.9%
Integrated Gradients	$1.54\times$	80.3%
Counterfactual (Flip Rate)	$2.46\times$	—
LIME	$2.87\times$	95.3%

† Only method that *disagrees* with the others.

Table 4: Drug/non-drug token importance ratio across all five XAI methods. SHAP, LIME, and IG computed on all 6,821 sentences; counterfactual on 6,821; attention on 50.

All four non-attention methods agree that drug tokens are disproportionately important. SHAP ($1.23\times$) and IG ($1.54\times$) show moderate drug token dominance across the full 6,821-sentence corpus, providing strong convergent validity. LIME ($2.87\times$) shows the strongest signal, consistent with its perturbation-based approach that samples a broader neighborhood of the decision boundary. Counterfactual flip rates ($2.46\times$) measure causal necessity directly and fall between gradient-based and LIME estimates.

6.6 Brand Substitution: Lexical Brittleness

A 13.8% flip rate from brand-for-generic substitution (H2 confirmed) reveals that the model over-indexed on specific generic drug strings during training. When those strings are replaced by clinically

Comprehensive XAI Analysis — BioBERT ADE Detection

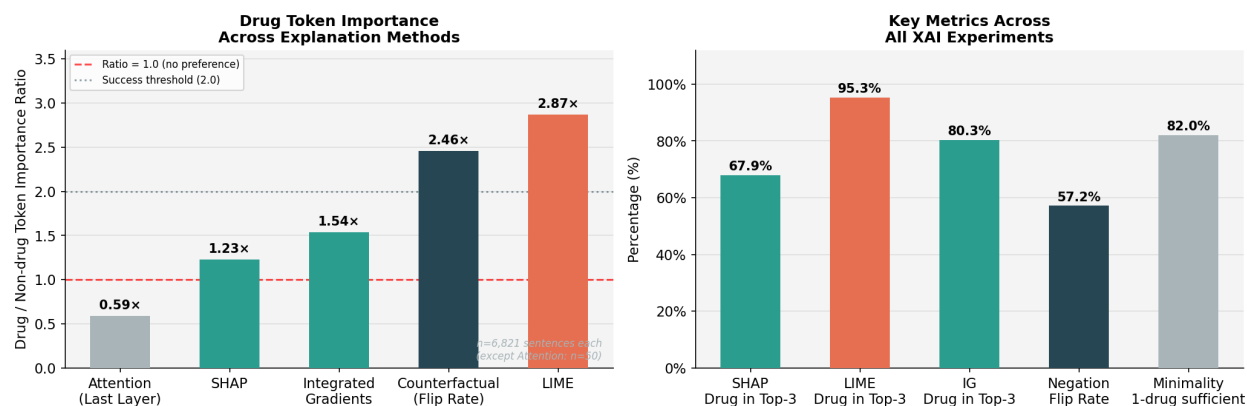


Figure 3: Left: drug/non-drug token importance ratio across all five XAI methods. Attention (0.59x) is the sole outlier. Right: key metrics across all experiments.

identical brand equivalents (e.g., *naproxen* → *Vi-movo*), 1 in 7 ADE-positive predictions changes — a direct deployment risk for clinical notes where brand names predominate.

6.7 Counterfactual Minimality

Of 200 flip sentences analyzed, 164 (82.0%) flip with just one drug token masked; 28 (14.0%) require two; 6 (3.0%) require three. Drug-token reliance is highly concentrated, not distributed. Even in multi-drug sentences, one token is sufficient in 4 out of 5 cases.

6.8 Negation Counterfactuals

Table 5 reports flip rates by negation pattern.

Pattern	Flip%	n
presented with	100.0	3
showed	100.0	1
developed	74.5	55
experienced	66.7	3
induced	59.1	44
caused	57.1	7
associated with	38.9	18
following	20.0	5
reported	7.1	14
Overall	57.2	152

Table 5: Negation counterfactual flip rates by linguistic pattern (300 sampled; 152 matched a pattern). Low flip rate = model ignores that negation.

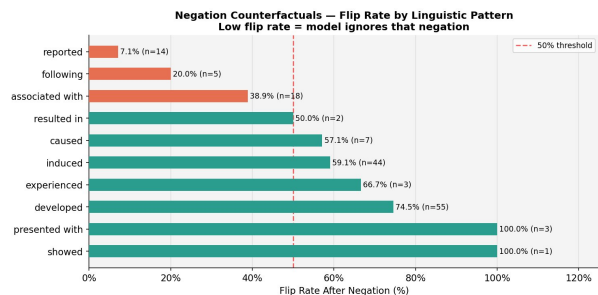


Figure 4: Negation counterfactual flip rates by linguistic pattern. Coral bars (<50%) indicate patterns the model mostly ignores.

42.8% of sentences are completely unaffected by negation insertion. The *reported* pattern is the most dangerous: only 7.1% flip rate, meaning “Patient did not report any ADE” is functionally equivalent to “Patient reported an ADE” for the model. Stronger causal verbs (*developed*: 74.5%, *induced*: 59.1%) are partially respected; weaker expressions are almost entirely ignored.

6.9 SHAP Interaction Analysis

Drug and context tokens push predictions in the same direction (cooperatively) in 88.6% of paired cases. Mean drug token SHAP is +0.129 vs. context SHAP magnitude of 0.135 — comparable in scale, but minimality analysis (82% single-token sufficient) shows context tokens *cannot substitute* for drug tokens. The cooperation is asymmetric.

Error Pattern Analysis — Family A (Drug Masking, ADE Sentences)

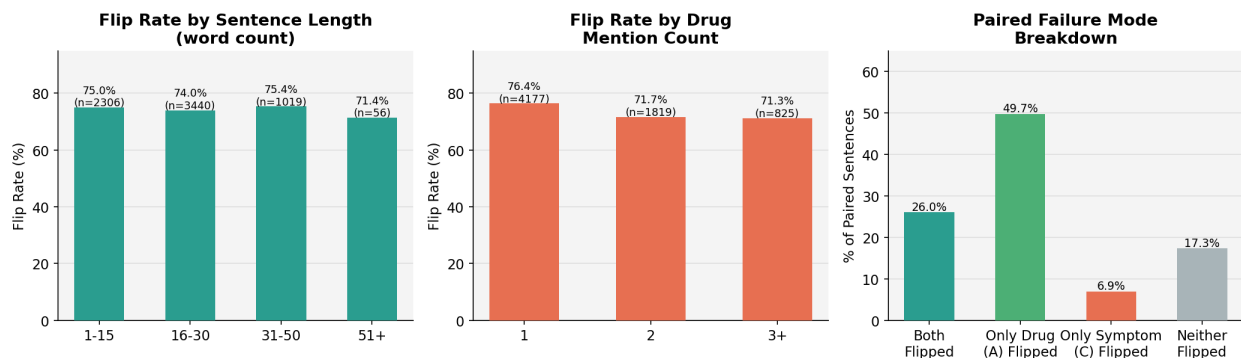


Figure 5: Error pattern analysis for Family A: flip rates by sentence length (left), drug mention count (centre), and paired failure mode breakdown (right).

7 Error Pattern Analysis

By sentence length. Flip rates are flat across lengths: 75.0% (1–15 words), 74.0% (16–30), 75.4% (31–50), 71.4% (51+). Longer sentences with richer clinical context do not reduce drug-token dependence.

By drug mention count. Single drug: 76.4%; two drugs: 71.7%; three or more: 71.3%. A slight redundancy effect is present but flip rates remain above 71% regardless.

By causal marker. Sentences with explicit causal language (“induced,” “caused by,” etc.) flip at 74.5%; without: 74.6%. The model is not using linguistic causal markers as a signal.

Paired failure modes. Among 4,270 paired sentences (present in both Family A and C): 49.7% flip only under drug masking (drug tokens are uniquely causal), 26.0% flip under both, 17.3% flip under neither (model relies on structure or co-occurrence), and only 6.9% flip only under symptom masking.

8 Discussion

Our multi-method analysis provides the strongest evidence to date that BioBERT’s ADE predictions are primarily driven by drug-name surface tokens. Four independent XAI methods (counterfactual, SHAP, LIME, IG) agree on direction while differing in magnitude — a pattern consistent with method-specific approximation biases rather than disagreement about the underlying causal structure.

The attention finding is the most practically significant for the XAI community: a method routinely used to *explain* clinical NLP predictions to clinicians points in the opposite direction from all other evidence. Deploying attention-based explanation interfaces in clinical settings would systematically mislead practitioners about why predictions were made.

The negation finding is the most practically significant for patient safety. A system that ignores explicit negation in 42.8% of cases cannot be safely deployed in clinical settings without explicit negation robustness evaluation. The 7.1% flip rate for the *reported* pattern — the most common reporting verb in clinical case reports — is particularly concerning.

SHAP interaction analysis (88.6% cooperative rate) suggests the model has learned a joint drug-context signal rather than a pure drug-name shortcut. However, minimality (82% single-token sufficient) shows this cooperation is asymmetric: context reinforces drug signal but cannot replace it.

9 Clinical Implications

ADE detection models intended for clinical deployment should be evaluated on counterfactual sensitivity diagnostics and negation robustness benchmarks

in addition to held-out F1. The Sensitivity Ratio ($2.46\times$) and negation sensitivity (57.2%) should be treated as model capability metrics alongside accuracy. Attention-based explanation interfaces should not be used without validation against causal methods. Brand-name augmentation during training is a concrete and low-cost remediation for the lexical brittleness finding.

10 Limitations

Attribution sample size. SHAP, LIME, and IG were computed on all 6,821 ADE-positive sentences, matching the counterfactual sample. SHAP interaction analysis was computed on 10 sentences due to the higher per-sentence cost of pairwise Shapley interactions. Attention analysis is limited to 50 sentences.

Single model. All findings are BioBERT-specific. Generalization to PubMedBERT, ClinicalBERT, or GPT-based models is an open empirical question.

RxNorm coverage. Only 51.5% of drug names mapped to brand equivalents, excluding obscure or off-patent drugs. The brittleness finding likely understates the true scale.

Negation pattern coverage. Only 152 of 300 sampled sentences matched one of 10 negation patterns. Clinical negation is more varied, and true negation sensitivity may differ when evaluated across a broader rule set.

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Attention aggregation. We used last-layer attention averaged across heads. Different layer or head selections may yield different ratios. Our finding that attention disagrees with other methods may partially reflect this aggregation choice.

11 Conclusion

Using five complementary XAI methods on a fine-tuned BioBERT ADE detector, we find strong convergent evidence that the model disproportionately relies on drug-name surface tokens. The Sensitivity Ratio of 2.46 (counterfactual, $p < 0.001$), LIME ratio of $2.87\times$, SHAP ratio of $1.23\times$, and IG ratio of $1.54\times$ (all on 6,821 sentences) all independently confirm drug-token importance exceeds symptom-token importance. Attention weights ($0.59\times$) are the sole outlier, replicating [Jain and Wallace \(2019\)](#) in a clinical setting. Negation blindness (42.8% of sentences) and lexical brittleness (13.8% brand flip rate) represent concrete deployment risks. Minimality analysis reveals that drug-token reliance is highly concentrated: 82% of flip sentences require only one drug token to be masked. SHAP interaction analysis shows this drug-token dominance coexists with cooperative drug-context signals, suggesting partial but insufficient clinical reasoning. Together these findings call for counterfactual sensitivity and negation robustness as standard evaluation criteria for clinical NLP models.

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